EL-HOUCINE OUSSI

Software Engineer • Machine Learning Engineer

oussi.elhoucine0@gmail.com +86 177 5100 8739 (CN) / +212 642 587 839 (MA) Nanjing, China & Oujda, Morocco

oussi108.github.io github.com/Oussi108

EDUCATION

Nanjing University of Information Science and Technology (NUIST) — *Nanjing, China* Sep 2024 – Jun 2027
Master of Science in Computer Science and Technology
- **Research Focus:** Solar power & renewable energy forecasting, spatio-temporal deep learning, and atmospheric data modeling.
- **Core Work:** Multi-horizon prediction across 50 solar sites using satellite imagery (Himawari-8/9) and ECMWF ERA5 reanalysis datasets.

SUPMTI (Higher School of Management, Telecom & Computer Science) — *Oujda, Morocco* Sep 2021 – Jun 2024
State Engineering Degree in Computer Science & Information Systems
- **Core Focus:** Distributed Systems, Software Engineering, Clean Architecture, Cloud Integration, and Database Systems.

ISTA Lazaret (Specialized Institute of Applied Technology) — *Oujda, Morocco* Sep 2019 – Jun 2021
Specialized Technician Diploma in Software Development (TS)
- **Core Focus:** Object-Oriented Programming (OOP), Relational Databases (SQL Server/MySQL), Full-Stack Web Development.

TECHNICAL SKILLS

Languages:	Python, C# (.NET), Dart, Java, C/C++, TypeScript, JavaScript, SQL, Bash / Shell
Frameworks & ML:	ASP.NET Core 8 / MVC, Flutter, PyTorch, LightGBM, XGBoost, Scikit-learn, Django, Spring Boot, React
Databases & Cloud:	PostgreSQL, SQL Server, MySQL, SQLite, MongoDB, Firebase (Auth/Firestore), Docker, Git, Linux, IIS
Architecture:	Clean Architecture, RESTful APIs, MVVM (Riverpod), Microservices, Time-Series Modeling, CI/CD

WORK & RESEARCH EXPERIENCE

NUIST Renewable Energy AI Lab — Graduate AI & Data Science Researcher Sep 2024 – Present | Nanjing, China
- Engineered scalable benchmark pipelines for 50 commercial Australian solar farms comparing persistence, GBDT, and neural architectures.
- Processed multi-terabyte spatio-temporal datasets: Himawari-8/9 geostationary satellite channels and ERA5 numerical weather fields.
- Published open-source benchmark dataset & evaluation framework on Zenodo (DOI: 10.5281/zenodo.23590205) with 100% reproducible recipes.

Majal Berkane – Smart City — Backend & Mobile Developer Intern Mar 2024 – Jul 2024 | Berkane, Morocco
- Architected cross-platform mobile app in Flutter adopting MVVM and Riverpod for reactive state management.
- Engineered high-throughput REST APIs in ASP.NET Core following Clean Architecture principles and repository patterns.
- Optimized database queries and server-side caching, boosting IoT sensor telemetry retrieval speeds by over 35%.

Direction Provinciale de l'Équipement, du Transport et de la Logistique — Web Engineering Intern Mar 2023 – Jul 2023 | Oujda, Morocco
- Built an internal web platform in Django automating personnel, equipment scheduling, and administrative logistics workflows.
- Implemented RBAC authentication and automated report data processing pipelines using Pandas and NumPy.
- Configured and deployed production application on Windows Server with IIS and automated SQL Server backups.

SELECTED PROJECTS

PV Forecast Benchmark Python PyTorch LightGBM Zenodo [Code | DOI: 10.5281/zenodo.23590205] 2024 – Present
- End-to-end benchmark framework evaluating forecast horizons (15-min to day-ahead) across 50 solar generation sites.

Connect Check App Flutter Firebase Riverpod SQLite [GitHub] 2024 – 2025
- Real-time cross-platform app featuring live interaction sync, multi-language localization via Riverpod, and offline SQLite storage.

EcoBeach Environmental Monitoring Platform .NET Core 8 C# Clean Arch IoT 2024
- Clean Architecture backend monitoring coastal water quality, environmental IoT sensors, alert thresholds, and analytical reporting.

MystoGuess Mobile Game Flutter Dart Google Play [GitHub] 2024
- Published Android puzzle game on Google Play Store featuring dynamic animations, custom sound engines, and state persistence.

LANGUAGES

English: Fluent (Professional working & research) | **French:** Proficient (Professional working proficiency) | **Chinese (Mandarin):** Conversational (HSK 3) | **Arabic:** Native | **Tamazight:** Native